

Research interests: Social justice applications of data science, reduced-order modeling, data-driven modeling, machine learning, modal decompositions, transport-dominated physics, traveling waves, fluid dynamics

Education

University of Washington

2016-2021

Ph.D. in Mechanical Engineering, M.S. 2018

Seattle, WA

Thesis: Data-Driven Reduced-Order Modeling for Symmetrical Systems

Advisors: Steven Brunton (Mechanical Engineering) and J. Nathan Kutz (Applied Mathematics)

Seattle University

2012-2016

B.S. in Mechanical Engineering, Minor in Mathematics

Seattle, WA

Senior Design project with Ingersoll Rand, resulting in patent

Research advised by John Carter (Mathematics), supported by Clare Boothe Luce Foundation Undergraduate Research Award

Teaching

Instructor

DATA 5310 **Data Visualization** F23, F22
DATA 5901 **Capstone I** W25, W24, W23
DATA 5902 **Capstone II** S24, S23
DATA 5322 **Statistical Machine Learning II** S24, S23, S22
DATA 3310 **Data Visualization** W24, F22
MATH 1335 **Calculus II** W22, F21

Capstone Advisor

MSDS 25.2 **Janani Krishnamurthy, Satwik Kunaparaju, Ting-Yu Lin, Prasanna Nalabothu, Surya Ramesh**
Planning and Mission Effectiveness Demographic Data History, Archdiocese of Seattle
MSDS 25.5 **Lavanya Bunadri, Garrett Ringler, Ruqhaiya Syeda, Maelice Yamdjieu**
A Modern Approach to Pacific Salmon Research: An Online Dashboard, NOAA Fisheries
MSDS 24.3 **Wai Chun Chong, Malia Cortez, Tuan Anh Nguyen, Anna Tsai**
Meta-Analysis of Elwha River Water Temperature Regime, NOAA Fisheries, Fish Ecology Division
MSDS 24.5 **Yen-Ting Chen, Hanisha Karri, McKenzie Maidl, Justus Ngunjiri**
Washington State Craft Brewing Industry Snapshot, Washington Brewers Guild
MSDS 23.1 **Liya LaPierre, Karthika Selvaraj, Aishwarya Saibewar, and Radhika Magaji**
Donor Database Analysis and Recommendations, Camp Korey
MSDS 23.3 **Prateek Kakkar, Ramgopal Reddy Putta, and Xuanzhi Li**
A Data Curation System and Predictive Model for Unconfined Concrete Strength, DeSimone Consulting Engineers

Publications

In Progress

Community-Driven Data Science Practices

La Matematica

A Barreda II, C Diaz Eaton, S Hansen, JE Hibdon, L Gordon, R Greenwald, M José Gutiérrez Paz, K İnce, C Kelling, A Mendible, J Mercado

Under review 2023

Complete Data for Studying the Role of Demographics in Selection of the Nobel Prizes

A Mendible, J Higdon, M Nagy, R Roca, N Rodríguez, L Xian, C Topaz

Paper in preparation

Peer-Reviewed Journal Publications

- Data collection and analysis for small-town policing: challenges and recommendations** *Statistics and Public Policy*
C Kelling, A Haensch, **A Mendible**, S Brooks, A Wiedemann, M Aminian, W Hasty, J Higdon
DOI: 10.1080/2330443X.2024.2379270 2024
- Data science and social justice in the mathematics community** *Notices of the Am Math Soc*
Q Jones, AR Vindas Meléndes, **A Mendible**, M Aminian, HZ Brooks, N Alexander, C Diaz Eaton, P Chodrow
DOI:10.1090/noti2773 2023
- Data-driven modeling of two-dimensional detonation wave fronts** *Wave Motion*
A Mendible, W Lowrie, SL Brunton, JN Kutz
DOI: 10.1016/j.wavemoti.2021.102879 2022
- Data-driven modeling of rotating detonation waves** *Phys Rev Fluids*
A Mendible, J Koch, H Lange, SL Brunton, JN Kutz
DOI: 10.1103/PhysRevFluids.6.050507 2021
- Dimensionality reduction and reduced-order modeling for traveling wave physics** *Theor Comput Fluid Dyn*
A Mendible, SL Brunton, AY Aravkin, W Lowrie, JN Kutz
DOI: 10.1007/s00162-020-00529-9 2020
- Reduced strain mechanochemical activation onset in microstructured materials** *Polym Chem*
JJ Schwartz, R Behrou, B Cao, M Bassford, **A Mendible**, C Shaeffer, AJ Boydston, N Boechler
DOI: 10.1039/C9PY01875A 2020
- Randomized nonnegative matrix factorization** *Pattern Recognit Lett*
NB Erichson, **A Mendible**, S Wihlbom, JN Kutz
DOI: 10.1016/j.patrec.2018.01.007 2018

Professional Articles

- Counting on Justice: Bridging Quantitative Research and Community Empowerment** *MAA Focus*
A Mendible, C Kelling 2024
- Quantitative Justice: Using Data Science for Good** *MAA Math Values*
A Mendible 2024
- Getting Away From One Right Answer** *COMAP Consortium Newsletter*
A Mendible 2022

Patents

- Air Flow Hour Meter** *US Patent 10,713,858*
A Mendible, S Heard, T Tran, A Hardesty 2017

Presentations

- Quantitative Justice Through R: Data Provenance** *Upcoming 2025*
AMS Engaged Pedagogy Series Virtual
- Data Science Tools for Community-Driven Police Accountability** *Jan 2025*
NAM-SIAM-AMS Special Session on Quantitative Justice, Joint Mathematical Meetings Seattle, WA
- Data Science Tools for Community-Driven Police Accountability** *Jul 2024*
Quantitative Justice through Applied Data Science Minisymposium, SIAM Annual Meeting Spokane, WA
- Panel: Decoding Ethics: Perspectives on Responsible Data Science** *May 2024*
Women In Data Science Puget Sound Seattle, WA
- Small Town Police Accountability: Case Studies and Opportunities in Data Science** *Jan 2024*
NAM-SIAM-AMS Special Session on Quantitative Justice, Joint Mathematical Meetings San Francisco, CA
- Data Science for Justice: Small Town Policing** *Nov 2023*
QSIDE Datathon4Justice, Opening Plenary Virtual
- Small Town Police Accountability: A Data Science Toolkit** *Jul 2023*
SciPy: Scientific Computing with Python Austin, TX

Using Data Science for Social Justice	Mar 2023
Annual Meeting of the PNW Section of the Mathematical Association of America	Newberg, OR
QSIDE's SToPA (Small Town Policing Accountability) Research Lab	Oct 2022
QSIDE Datathon4Justice, Opening Plenary	Virtual
Meet Your Data Science Faculty	Oct 2022
Seattle University Mathematics Seminar	Seattle, WA
The Role of QSIDE's SToPA Research Lab in the Field of Criminal Justice	Apr 2022
QSIDE Data4Justice Conference	Virtual
Reduced-Order Modeling	Mar 2022
Pacific Lutheran University Mathematics Seminar	Virtual
Data-driven Modeling of Shock Waves in Rotating Detonation Engines	Mar 2021
SIAM Computational Science and Engineering	Virtual
Data-driven Modeling of Traveling Waves	Dec 2020
University of Delaware SIAM Seminar Series	Virtual
Data-driven Modeling of Traveling Waves	2020
Steve Brunton channel video abstract (14k views)	YouTube
Data-driven Modeling of Detonation Wave Interactions in Rotating Detonation Engines	Nov 2020
73rd Annual Meeting of the APS Division of Fluid Dynamics	Virtual
Dimensionality Reduction and Reduced Order Modeling for Traveling Wave Physics	Nov 2019
72nd Annual Meeting of the APS Division of Fluid Dynamics	Seattle, WA
Poster: The Space-Time Problem for Model Reduction of Traveling Waves	Jun 2019
DTRA Technical Program Review	Fairfax, VA
The Space-time Problem with Model Reduction for Traveling Waves	Mar 2019
SIAM Computational Science and Engineering	Spokane, WA

Scholarly Development

Grant Funding

Optimization of Cryoprotective Agents for Single-Mode Electromagnetic Resonant (SMER)	\$50,000
Rewarming Technology to Enable Long-Term Organ Preservation	2023
Seattle U College of Science and Engineering	
Multidisciplinary Research Program Award with Shen Ren (Mechanical Engineering)	
Small Town Police Accountability Lab	Workshop hosting
Banff International Research Station for Mathematical Innovation and Discovery	2022
Proposal accepted	

Workshops

Community Engaged Learning Faculty Fellowship Workshop	Sep 2023
Seattle U Center for Community Engagement, Funded Participant	Seattle, WA
Educating at the Intersection of Data Science and Social Justice Workshop	Jul 2023
Institute for Computational and Experimental Research in Mathematics, Brown University, Funded Participant	Providence, RI
Community Engaged Learning Workshop Series	May 2023
Seattle U Center for Community Engagement	Seattle, WA
Data Science and Social Justice: Networks, Policy, and Education Workshop	Jun-Jul 2022
Institute for Computational and Experimental Research in Mathematics, Brown University, Funded Participant	Providence, RI
Project NExT Fellow	2021-2022
Mathematical Association of America	Virtual
Preparing for Careers in Teaching Statistics and Data Science Workshop	Aug 2021
Section on Statistics and Data Science Education of the American Statistical Association	Virtual
EPSRC Summer School: Modal Decompositions in Fluid Mechanics	Aug 2019
Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Funded Participant	Cambridge, UK

Research Students

Emma Oriol

Redlining and Gentrification

AY23-24

MS Data Science

Alexander Churhill

Historical Redlining and Housing Quality in Seattle

AY23-24

MS Data Science

Chang He

Historical Redlining and Cycling Accessibility

AY23-24

MS Data Science

Malia Cortez

Historical Redlining and Population Flows

AY23-24

MS Data Science

Service

Academic Service

MS Data Science Student Admissions Review

2022-present

Review ~250 student applications per year, recommended admissions decisions

Chile Study Abroad Trip Co-Lead

2024

Assisted in supervising spring break student trip for Mathematical Models of Near-Shore Phenomena course

Math Department Community Building

Organize end-of-year student celebration for graduating MSDS cohort

2023, 2024

Organize weekly (AY 22-23) or quarterly (AY 23-24) activities and events to build departmental community

2022-2024

Data Science Faculty Hiring Committee

Reviewed 54 candidate applications, 14 screening interviews, hosted 4 all-day candidates, 2 hires

2023

Reviewed ~9 candidate applications, 9 screening interviews, hosted 2 all-day candidates, position unfilled

2022

Math Department Recruitment, Inclusion, Diversity, Equity and Retention Committee

2021-present

Member, led climate survey creation and distribution

QSIDE Institute Consortium SU Liaison

2021-present

Lead event coordination, advertisement at SU, and student involvement

Professional Service

QSIDE Data Science, Police Accountability and Community Engagement Research Lab

2021- present

Co-Director, lead meetings and organize research activities

SciPy Conference

Program Committee, organize topics and oversee tracks conference-wide

2025

Track Chair: Human Networks, Social Sciences, and Economics

2024

Women In Data Science Puget Sound

2024

Invited mentor for Mentorship Night career-seeking women

QSIDE Datathon4Justice

Co-Organizer

2023

Team Lead

2021, 2022, 2023

Humanities and Social Sciences Communications

Referee

-

Journal of Fluid Mechanics

Referee

-

Community Service

Mechanical Engineering Students Against Racism, UW Mechanical Engineering

2020-2021

Crafters Against COVID-19

2020

Engineering Discovery Days, UW College of Engineering

2016, 2019

Bike & Build Drift West

2017

UW Society of Women Engineers, Elementary school outreach visits

2016

Honors & Awards

Community Engaged Learning Faculty Fellowship , Seattle U Center for Community Engagement	AY 23-24
Supervisor of the Year , Seattle University Student Employment Shine Award	2023
Project NExT Fellow , Mathematical Association of America	2021
Graduate Diversity Fellowship , UW Graduate Opportunities and Minority Achievement Program, Full Support	2016, 2020
Steve and Lynn Pratt Fellowship , UW Mechanical Engineering	2017, 2020
Wynne Alexander Guy Spirit of the Mathematics Department Award , SU Mathematics	2019
Wei Li and Dongmei Chen Endowed Fellowship , UW Mechanical Engineering	2017
Brian and Mary Fabien Endowed Fellowship , UW Mechanical Engineering	2016, 2017
Undergraduate Research Award , Clare Boothe Luce Foundation	2014

Additional Experience

Clear Motion (formerly Levant Power)	June - Aug 2016
Controls Engineering Intern	Woburn, MA
National Renewable Energy Laboratory	June - Aug 2015
Science Undergraduate Laboratory Internship	Boulder, CO
Network for Earthquake Engineering Simulation	June - Aug 2014
OH Hinsdale Wave Research Laboratory REU	Corvallis, OR